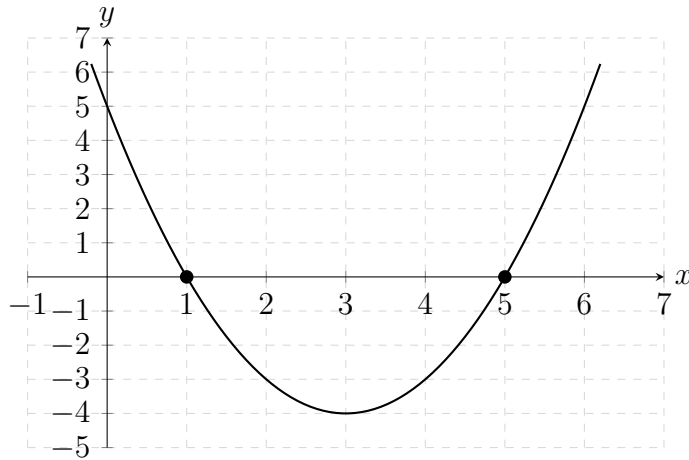


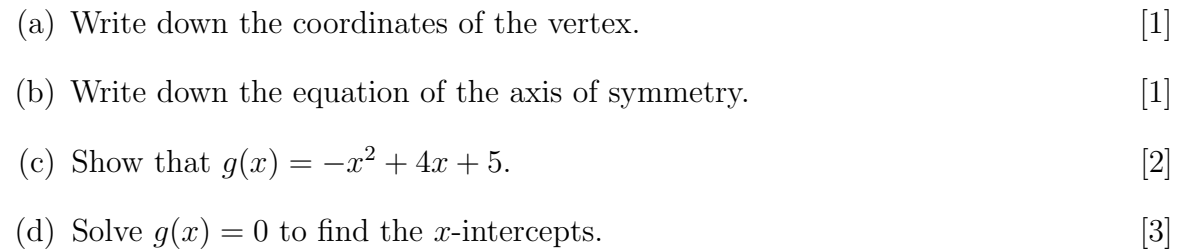
6.5 Classwork: Quadratic Functions and Symmetry

1. The diagram shows the graph of a quadratic function with x -intercepts at $(1, 0)$ and $(5, 0)$.

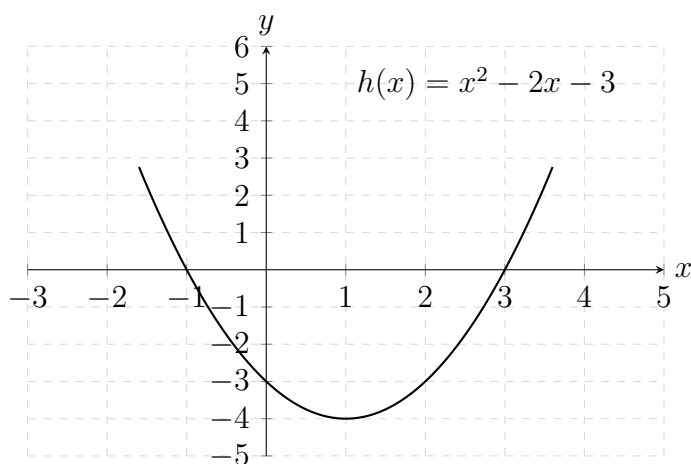


- (a) Find the equation of the axis of symmetry. [1]
- (b) Write the equation of the parabola in factored form, $y = (x - p)(x - q)$. [1]
- (c) Expand to show that the equation is $y = x^2 - 6x + 5$. [2]
- (d) Find the coordinates of the vertex. [2]
- (e) Write down the y -intercept. [1]

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3. The diagram shows part of the graph of $h(x) = x^2 - 2x - 3$.



- (a) Write $h(x)$ in factored form, $(x - p)(x - q)$. [2]
- (b) Write down the x -intercepts of the parabola. [1]
- (c) Find the equation of the axis of symmetry. [1]
- (d) Find the coordinates of the vertex. [2]
- (e) Write $h(x)$ in vertex form, $(x - h)^2 + k$. [1]

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4. Consider the quadratic function $p(x) = -x(x - 4)$ with domain $-1 \leq x \leq 5$.

(a) Write down the x -intercepts. [1]

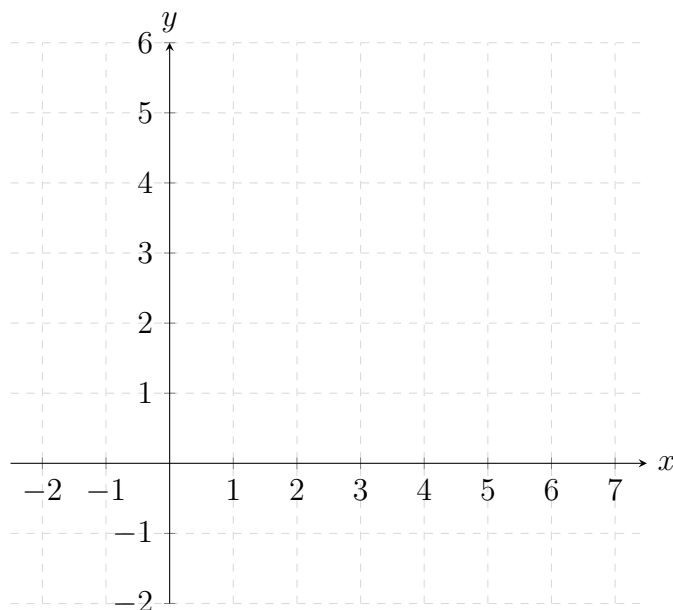
(b) Find the equation of the axis of symmetry. [1]

(c) Find the coordinates of the vertex. [2]

(d) Expand to write $p(x)$ in the form $ax^2 + bx + c$. [1]

(e) On the grid below, sketch the graph of $p(x)$. Label the vertex and x -intercepts. [2]

(f) Write down the range of the function over the given domain. [1]



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